THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 11: **The World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An algorithm is best described as: 1

(a) a calculator button
(b) a single number obtained by guessing
(c) a precise step-by-step procedure that solves a problem
(d) a table of answers learnt by heart

2. In the addition algorithm the digits are aligned from the right because: 1

(a) units must be added to units, tens to tens, and so on
(b) we read numbers from right to left
(c) carries move to the right
(d) the right digit is the largest

3. Euclid's subtraction algorithm is based on the fact that for m ≥ n, gcd(m, n) equals: 1

(a) gcd(n, m − n) **(b)** gcd(m − n, m + n)
(c) gcd(m, m + n) **(d)** gcd(m × n, n)

4. **Assertion (A):** Euclid's subtraction algorithm finds gcd(375, 825) = 75. 1
Reason (R): Every common divisor of m and n is also a common divisor of n and m − n, and conversely.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

5. While adding 478 and 385 by the column algorithm, the number of columns that produce a carry of 1 is: 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** 3

6. While adding 7777 and 3333 by the column algorithm, the number of columns that produce a carry of 1 is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 3

7. By Euclid's algorithm, gcd(252, 198) is: 1

(a) 36 **(b)** 18 **(c)** 19 **(d)** 9

8. The number of times Step 3 of Euclid's subtraction algorithm is applied while computing gcd(35, 10) is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 7

9. The list of divisors of 18 produced by the divisor algorithm has how many entries? 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

10. The prime divisors of 60 (from divisors(60) = [1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60]) are: 1

(a) 2, 3, 4, 5 **(b)** 2, 3, 5 **(c)** 2, 3, 5, 6 **(d)** 1, 2, 3, 5

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Define an algorithm and give one example from this chapter. 2

12. Add 5987 and 4376 using the column algorithm and state in which columns a carry is generated. 2

13. Why is the final carry written in Step 5 of the addition algorithm? Give an example in which omitting it gives a wrong answer. 2

14. Execute the divisor algorithm for 20 and write the final list. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Add 7968 and 5479 following the steps of the algorithm precisely, showing the carry in each column. 3

16. What happens when a 5-digit number is added to a 3-digit number by this algorithm? Explain how the steps handle it. 3

17. Explain why the column-addition algorithm gives the correct sum. 3

OR

Compare the time to add 3347 + 2729 by counting dots (1 dot per second) with the time by the column method (15 seconds per single-digit addition with carry). 3

18. Execute the divisor algorithm for 28 and write list-of-divisors after j = 1, 2, 4, 7 and at the end. 3

OR

Find the common divisors of 36 and 60 by the algorithm of Section 11.2.3 and hence gcd(36, 60). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Write the algorithm to add two numbers digit by digit. Execute it for 6785 + 2748, showing the carry in each column, and state why Step 5 is needed. 5

OR

Write the algorithm for divisors(n). Execute it for n = 24 and write the list. Why is the list in increasing order? 5

20. Write the complete first algorithm for gcd(m, n) using lists of divisors. Execute it for gcd(30, 45). 5

OR

Describe the three successive improvements to the first gcd algorithm (Section 11.3.1) and how many numbers are checked after each for m = 36, n = 60. 5

END OF QUESTION PAPER — Total: 40 Marks